

**Title : Creating VLAN and VLAN Trunking using Packet Tracer**

Theory:

A VLAN (Virtual Local Area Network) is a network configuration that divides a single physical network into multiple logical networks. Each VLAN operates as an independent network, even if multiple VLANs share the same physical infrastructure. VLANs enhance network security, minimize broadcast traffic, and allow administrators to segment traffic logically often based on departments, functions, or user groups within an organization.

VLAN Trunking

VLAN trunking is a technique that allows traffic from multiple VLANs to pass through a single network link between switches or other network devices. This is achieved by tagging Ethernet frames with a VLAN identifier, most commonly using the IEEE 802.1Q standard. Trunking enables VLANs to extend across multiple devices and locations, providing greater flexibility in network design and management.

VLAN Architecture

VLAN architecture logically groups devices into separate broadcast domains within the same physical network. Each VLAN typically represents a different logical network, and communication between VLANs is restricted unless routing or firewall rules are configured.

Key components include:

- Access ports – Connect end devices to a specific VLAN.
- Trunk ports – Carry traffic for multiple VLANs between devices.
- VLAN aware devices – Switches and routers that manage VLAN traffic.

This architecture improves scalability, security, and performance in modern networks.

Components Used

Hardware: 2 Switches, Ethernet cables, 4 End devices

Software: Cisco Packet Tracer

Network Diagram

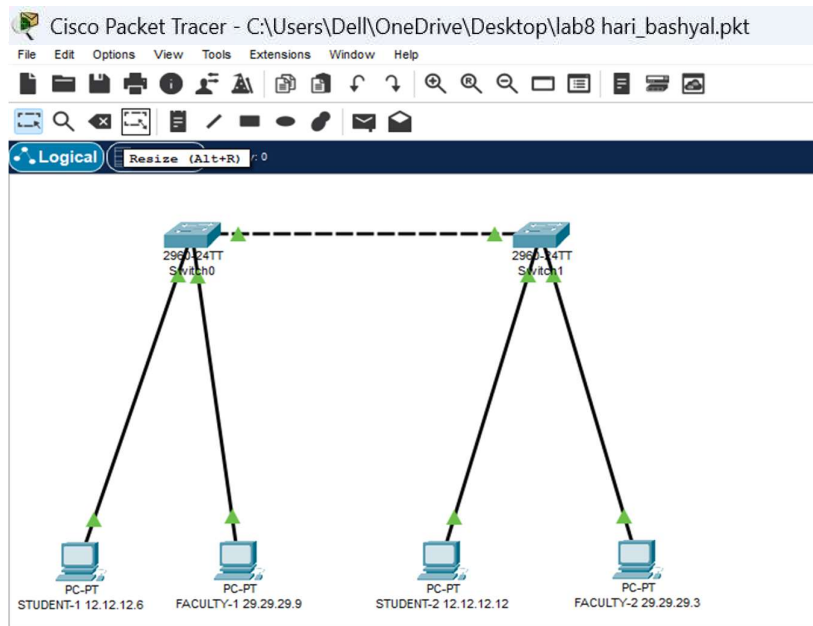


Fig: Network map for VLAN

Procedure

The following steps describe how to create the LAN network shown in the diagram using Cisco Packet Tracer:

Step 1: Launch Cisco Packet Trace

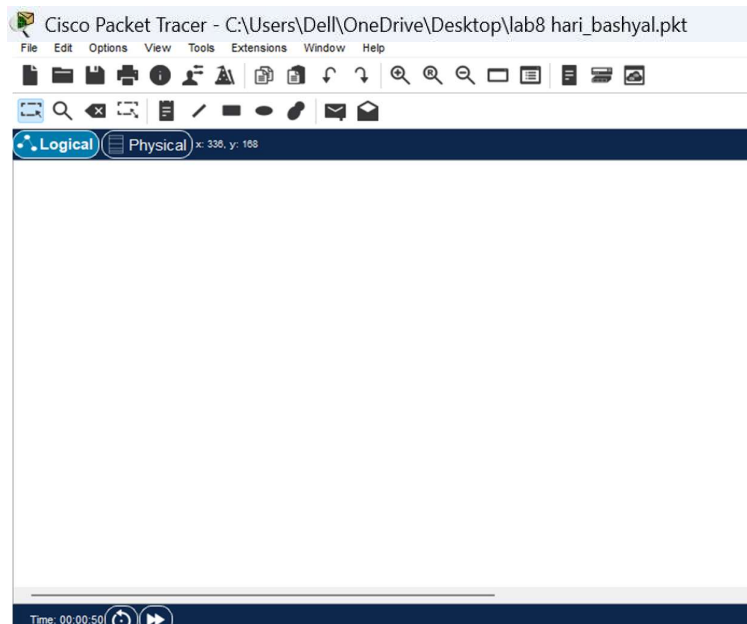


Fig: Workshop for network design

Step 2: Add Network Devices to the Workspace

- From the Device-Type Selection Panel, choose the following devices and place them in the workspace:
 - Two 2960-24TT Switch
 - Four PCs (labeled STUDENT-1 , STUDENT -2 , FACULTY-1 AND 2)

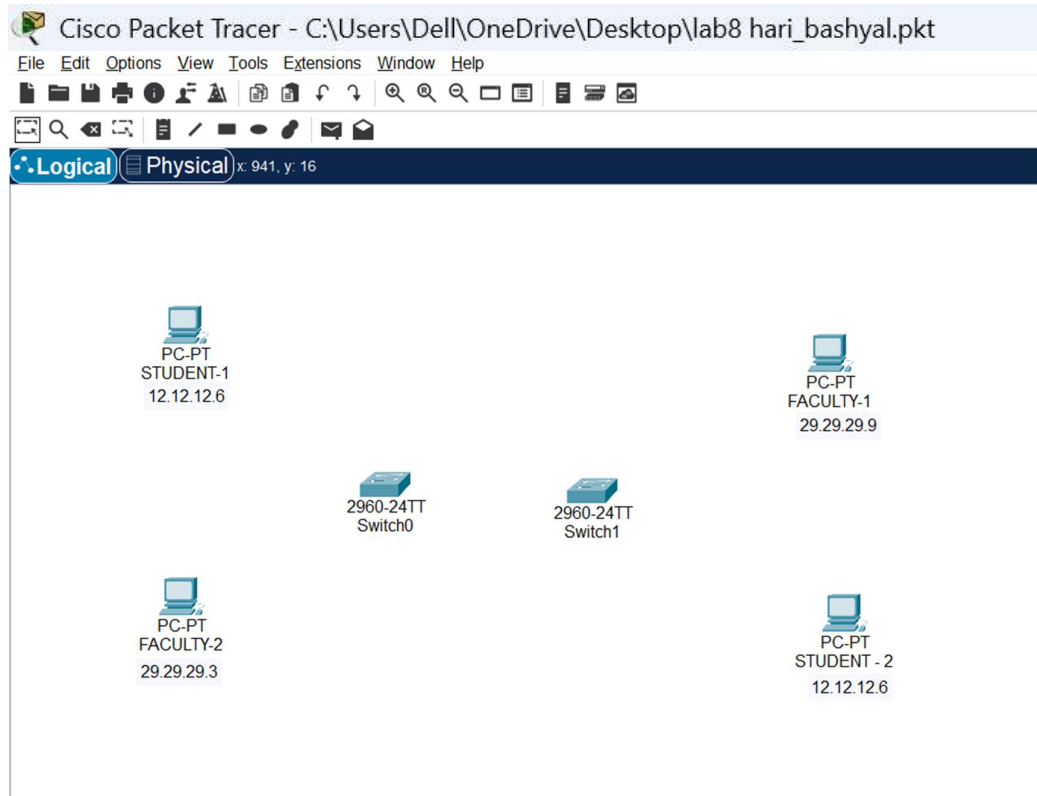


Fig: Switch and PCs for LAN creation

Step 3: Connect the Devices

- Select the Copper Straight-Through Cable from the connection options.
- Connect each PC to an available FastEthernet port on the switch.
- Ensure all connections are properly established, and link lights on the devices turn green to indicate connectivity.

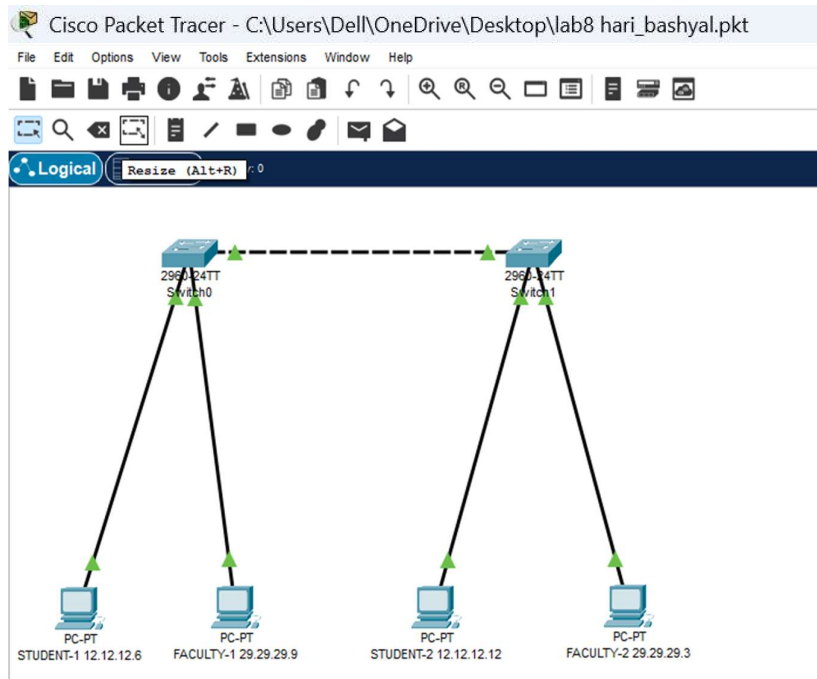


Fig: Connection between all devices in VLAN

Step 4: Configure IP Addresses

- Right-click on each PC in the workspace and select IP Configuration from the Desktop tap.
- In the IP Configuration window, enter the IP address as and subnet mask for each PC.

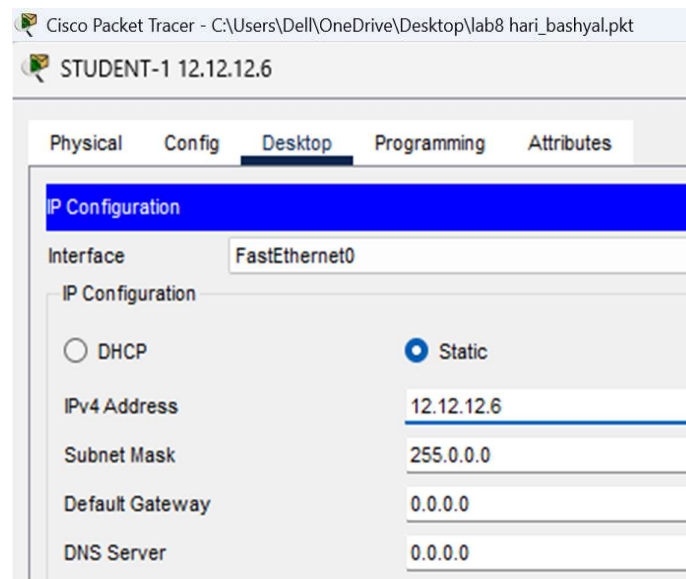


Fig: IP Configuration

Step 5: Configuring VLANs

- Create VLAN on both Switches & Assign Port to both switches.
- Create trunkation in both switches.

Code for VLAN configurations:

```
Switch(config)#vlan 10
Switch(config-vlan)#name student
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name faculty
Switch(config-vlan)#exit
Switch(config)#exit
```

Code for Assigning ports:

```
Switch#config t
Switch(config)#int fa 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#int fa 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#exit
```

Code for Trunking Switches: Switch#config t

```
Switch(config)#int fa 0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
```

```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab8 hari_bashyal.pkt
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	student	active	Fa0/1
20	faculty	active	Fa0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Fig: Assigning ports to VLAN

```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab8 hari_bashyal.pkt
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa 0/3
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG I: Configured from console by console
```

Fig: Configuring trunking between switches

Step 6: Testing and Validation

1. To verify connectivity, use the ping command from one PC to another.
2. On a PC, open the Command Prompt from the Desktop tab.
3. Type: ping <IP address of the other PC>
4. If the ping is successful, you will receive replies from the destination PC, confirming that the network is functioning correctly.

The screenshot shows the Cisco Packet Tracer application window. The title bar reads "Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab8 hari_bashyal.pkt". Below the title bar, the device name "STUDENT-2 12.12.12.12" is displayed. The "Desktop" tab is selected in the top navigation bar. The "Command Prompt" window is open, showing the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 12.12.12.6

Pinging 12.12.12.6 with 32 bytes of data:

Reply from 12.12.12.6: bytes=32 time=3ms TTL=128
Reply from 12.12.12.6: bytes=32 time<1ms TTL=128
Reply from 12.12.12.6: bytes=32 time<1ms TTL=128
Reply from 12.12.12.6: bytes=32 time<1ms TTL=128

Ping statistics for 12.12.12.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

Fig: Successful connectivity test between STUDENT-1 and STUDENT-2

The screenshot shows the Cisco Packet Tracer application window. The title bar reads "Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab8 hari_bashyal.pkt". Below the title bar, the device name "FACULTY-2 29.29.29.3" is displayed. The "Desktop" tab is selected in the top navigation bar. The "Command Prompt" window is open, showing the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 29.29.29.9

Pinging 29.29.29.9 with 32 bytes of data:

Reply from 29.29.29.9: bytes=32 time<1ms TTL=128
Reply from 29.29.29.9: bytes=32 time<1ms TTL=128
Reply from 29.29.29.9: bytes=32 time=1ms TTL=128
Reply from 29.29.29.9: bytes=32 time=3ms TTL=128

Ping statistics for 29.29.29.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

Fig: Successful connectivity test between FACULTY-1 and FACULTY-2

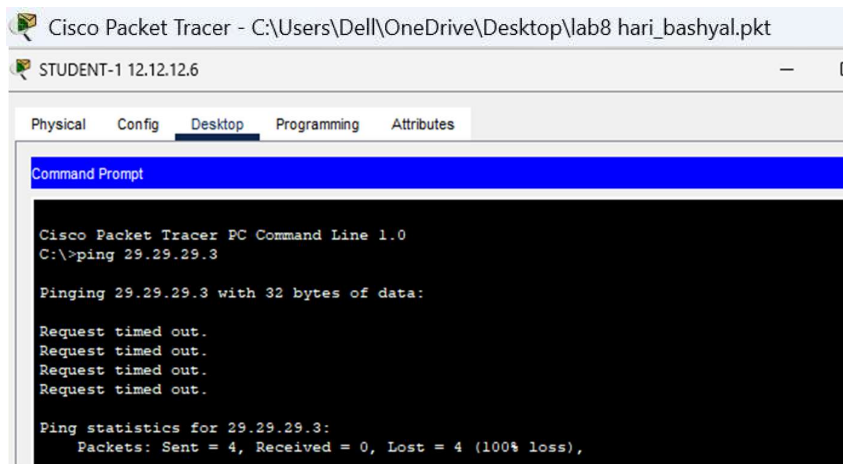


Fig: Connectivity test between STUDENT-1 to FACULTY-2

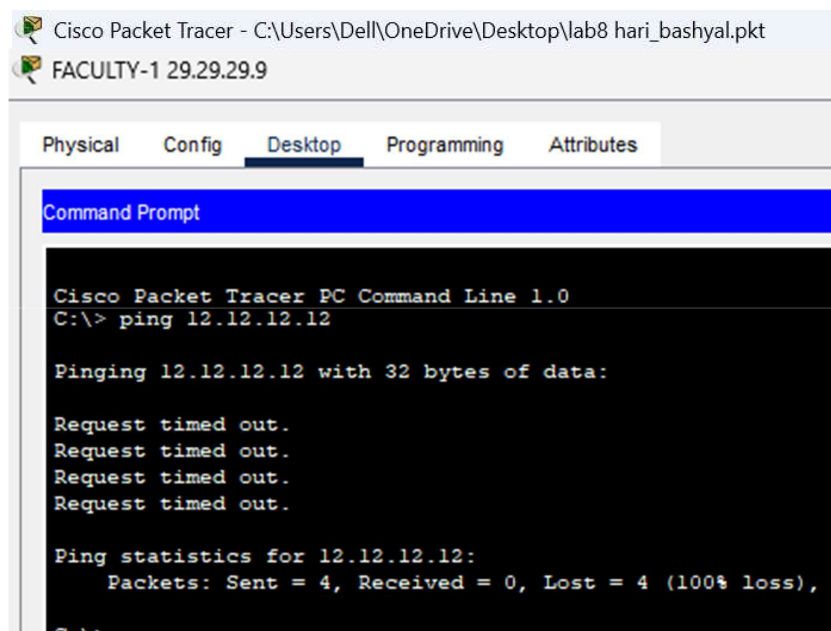


Fig: Connectivity test between FACULTY-1 to STUDENT-2

Addressing Table:

The addressing table is as follows:

Device	Interface	IPv4 Address	Subnet
STUDENT-1	NIC	12.12.12.6	255.0.0.0
STUDENT-2	NIC	12.12.12.12	255.0.0.0
FACULTY-1	NIC	29.29.29.9	255.0.0.0
FACULTY-2	NIC	29.29.29.3	255.0.0.0

CONCLUSION

In this lab, we worked on setting up VLANs and trunking in Cisco Packet Tracer to make our network more organized and efficient. By grouping devices into different VLANs, we reduced unnecessary broadcast traffic, improved security, and made the overall network flow cleaner. The trunk link, which used 802.1Q tagging, allowed multiple VLANs to travel over a single cable while still keeping each one separate. Altogether, this lab showed how proper planning and structured design can make a network easier to manage, scale, and keep secure.